



Lyall's Wren Journal

Ecosystem perturbations

C. Didon

Contents

1	Some definitions	1
1.1	Ecosystem	1
1.2	Perturbation	1
2	Types of perturbations	2
3	Perturbations, public perception and ecology reality	3

1 Some definitions

For the future explanation, I'll need to define the terms I'll use.

1.1 Ecosystem

An **ecosystem** (or ecological system) consists of all the organisms and the abiotic pools (or physical environment) with which they interact (Tansley 2016). It's all the living organisms (biocenosis) in an environment (biotope), with all the interactions between each other and with the environment. This includes therefore the energy flux between the organisms, the life cycle with the decomposition phase and the perturbations and resilience, which is the main topic of this article.

Another very important aspect of the ecosystem is its dynamic aspect. All the interactions are regulated in a certain way where the ecosystem stays generally stable. This introduces the concepts of perturbations (that alter that balance), resistance (face to perturbations) and resilience (after a perturbation) (Folke et al. 2004) (Grimm and Wissel 1997).

The ecosystem also lead to the concept of ecotope, the smallest ecosystem unit (Margulis and Fester 1991). See that this definition doesn't give a scale, in fact, a human can be considered as an ecosystem where the human is the environment and all the bacteria and microorganismes in it, are the organisms in interaction. This introduces the concept of meta-organisms also known as holobiont (Baedke, Fábregas-Tejeda, and Nieves Delgado 2020) (Tansley 2016), but this is a completely other subject.

1.2 Perturbation

In English, the disturbance designates the cause of a perturbation, which is the consequence.

A **disturbance** is "a cause; a physical force, agent, or process, either abiotic or biotic, causing a perturbation (which includes stress) in an ecological component or system; relative to a specified reference state and system; defined by specific characteristics".

A **perturbation** is "an effect; the response of an ecological component or system to disturbance or other ecological process as indicated by deviations in the values describing the properties of the

component or system; relative to a specified reference condition; characterized by direction, magnitude, and persistence". This can be temporary (back to previous state) or permanent (no return to previous state).

A **stress** is "an effect; a physiological or functional effect; the physiological response of an individual, or the functional response of a system caused by disturbance or other ecological process; relative to a specified reference condition; characterized by direction, magnitude, and persistence; a type of perturbation". (Rykiel 1985)

This definition of perturbation talk about a change, which necessitates the comparison to a previous state in the studied ecosystem. This state is usually the climax of the ecosystem (the state where mainly the biomass is maximal, the ratio productivity/biomass is the highest, maximum specific diversity).

There is also more characterisation of a perturbation than "direction, magnitude and persistence", such as size, duration, resistance and resilience. Let's define those terms too.

The **size** is simply the surface of the ecosystem impacted. It can be measured as a value (squared kilometers), or a ratio of the ecosystem surface.

The **duration** is the time from the disturbance to the next ecosystem climax (previous or new one).

The **intensity** is the physical force of the disturbance by surface unit and time.

The **severity** is the impact of the disturbance on organisms. (Chauchard 2025)

The **resistance** is the capacity to remain unchanged faced with a disturbance (Grimm and Wissel 1997).

The **resilience** is "the capacity of a system to absorb disturbance and reorganize while undergoing change so as to still retain essentially the same function, structure, identity, and feedbacks" (Walker et al. 2004).

The **persistence** is the capacity to respond face to a perturbation disturbance, to reach back the climax (previous or new one) (Holling 1973).

The **net change** is the difference between the climax before and after the disturbance.

Disturbances not happen only one time, so we can create a disturbance regimes concept. A **disturbance regime** is the evolution and characterisation of disturbances in time. This is characterised with occurrence, return interval, spatial organisation and specificity.

The **occurrence** is the number of disturbances in a given time and a given place.

The **return interval** is the duration between two disturbances.

The **spatial organisation** is the size and spacing of the disturbances.

The **specificity** is the selection of the agent between one or different habitats, species... (Chauchard 2025)

2 Types of perturbations

The disturbances can be regrouped in 3 main types.

- The **press** is a disturbance constantly increasing in strength along time. we can cite for instance the urbanization.
- The **pulse** is basically a disturbance in a gauss form. There is a huge disturbance very for a relatively short time. For instance wildfires.
- The **ramp** is a progression aggravation until it reaches a break point. For instance droughts.

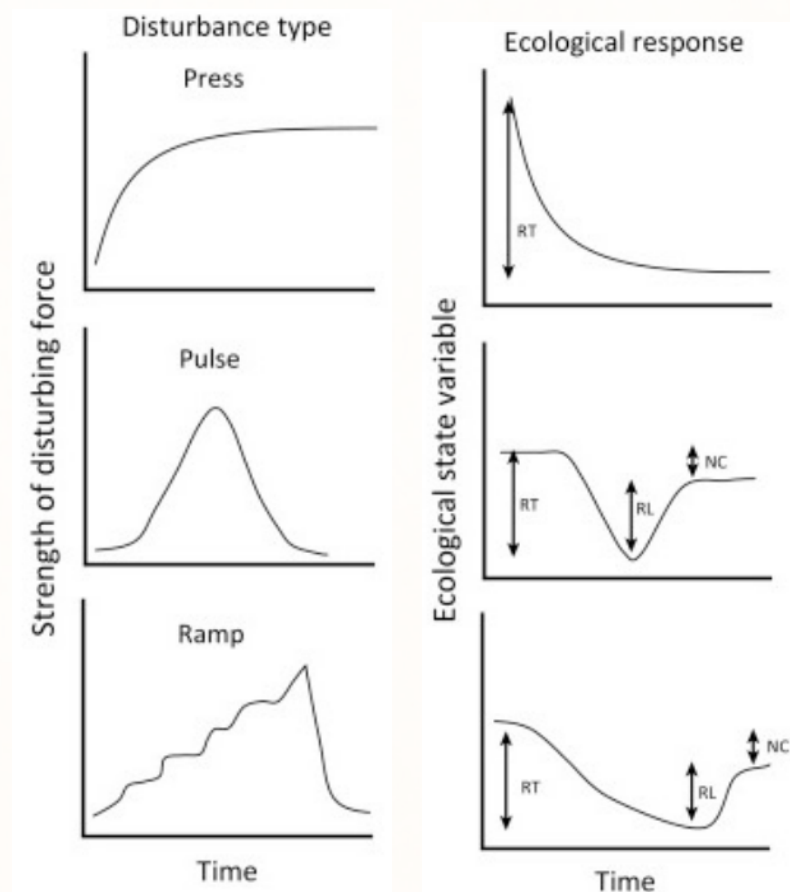


Figure 1: Disturbance types (Nimmo et al. 2015)
RT: resistance, RL: resilience and NC: net change

Note that these models exist in order to simplify, classify and explain disturbances. The reality is much more complex.

We can also categorize them in anthropic and non anthropic perturbations (anthropic is human induced). This can be useful to study the impact of the human species on the ecosystems.

3 Perturbations, public perception and ecology reality

Perturbations are often perceived as harmful events. This is not false but this is not true neither, as always the reality is way more complex. Some ecosystems are dependent of some perturbations, and this for different reasons. One species can be specifically adapted to this ecosystem where the perturbation occurs and need it to persist, or the perturbation allow a slightly different fauna to develop by killing the predominant specie (same dynamic that we see with keystone species) (this will be sourced in the future articles for each perturbations).

If i take the example of forest fire, it undeniably destroys plants, some habitats, kill some individuals. But the fire allow some different plant to grow after the perturbation for various reasons such as heat resistance, heat affinity, or just place to grow. The forest fire then allow a new ecosystem to exist with its own flora, fauna, interaction and regulation.

However, the anthropic perturbations are harmful, for instance global warming, urbanisation, deforestation, which are also press types and aren't reversible. This leads to the theorization of an Anthropocene, a geological era fundamentally altered by humans. (Lewis and Maslin 2015)(Dirzo et al. 2014)

This article is an introduction to the futur articles, that will be centred on a dynamics of perturbation, and the flora and fauna adapted to it.

References

- Baedke, Jan, Alejandro Fábregas-Tejeda, and Abigail Nieves Delgado (2020). “The Holobiont Concept before Margulis”. In: *Journal of Experimental Zoology Part B: Molecular and Developmental Evolution* 334.3, pp. 149–155. ISSN: 1552-5015. DOI: 10.1002/jez.b.22931. (Visited on 01/11/2026).
- Chauchard, S. (2025). *Chauchard’s lecture*.
- Dirzo, Rodolfo et al. (2014). “Defaunation in the Anthropocene”. In: *Science* 345.6195, pp. 401–406. ISSN: 0036-8075. JSTOR: 24745121. (Visited on 01/28/2026).
- Folke, Carl et al. (Dec. 2004). “Regime Shifts, Resilience, and Biodiversity in Ecosystem Management”. In: *Annual Review of Ecology, Evolution, and Systematics* 35. Volume 35, 2004, pp. 557–581. ISSN: 1543-592X, 1545-2069. DOI: 10.1146/annurev.ecolsys.35.021103.105711. (Visited on 01/11/2026).
- Grimm, V. and Christian Wissel (Feb. 1997). “Babel, or the Ecological Stability Discussions: An Inventory and Analysis of Terminology and a Guide for Avoiding Confusion”. In: *Oecologia* 109.3, pp. 323–334. ISSN: 1432-1939. DOI: 10.1007/s004420050090. (Visited on 01/11/2026).
- Holling, C. S. (Nov. 1973). “Resilience and Stability of Ecological Systems”. In: *Annual Review of Ecology, Evolution, and Systematics* 4. Volume 4, 1973, pp. 1–23. ISSN: 1543-592X, 1545-2069. DOI: 10.1146/annurev.es.04.110173.000245. (Visited on 01/11/2026).
- Lewis, Simon L. and Mark A. Maslin (Mar. 2015). “Defining the Anthropocene”. In: *Nature* 519.7542, pp. 171–180. ISSN: 1476-4687. DOI: 10.1038/nature14258. (Visited on 01/28/2026).
- Margulis, Lynn and René Fester, eds. (June 1991). *Symbiosis as a Source of Evolutionary Innovation: Speciation and Morphogenesis*. Cambridge, MA, USA: MIT Press. ISBN: 978-0-262-13269-5.
- Nimmo, D. G. et al. (Sept. 2015). “Vive La Résistance: Reviving Resistance for 21st Century Conservation”. In: *Trends in Ecology & Evolution* 30.9, pp. 516–523. ISSN: 1872-8383. DOI: 10.1016/j.tree.2015.07.008.
- Rykiel, Edward (Sept. 1985). “Toward a Definition of Ecological Disturbance”. In: *Australian Journal of Ecology* 10, pp. 361–365. DOI: 10.1111/j.1442-9993.1985.tb00897.x.
- Tansley, A. G. (Oct. 2016). *The Use and Abuse of Vegetational Concepts and Terms*. <https://web.archive.org/web/2016/content/uploads/2015/12/tansley-1935.pdf>. (Visited on 01/11/2026).
- Walker, Brian et al. (Sept. 2004). “Resilience, Adaptability and Transformability in Social–Ecological Systems”. In: *Ecology and Society* 9.2. ISSN: 1708-3087. DOI: 10.5751/ES-00650-090205. (Visited on 01/11/2026).